Mukul Bhutani

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Experience

Google — *Mountain View, CA*

Research Engineer, Google DeepMind — *February 2022 – Present*

- Core contributor to **Veo 3**, Google Deepmind's state of the art video generation model.
- Worked on making sure Veo 3 is developed in a safe and responsible manner.
- Core contributor to **Imagen** series, Google Deepmind's state of the art image generation models. Worked across the stack on data, evals, and post-training of the model.
- Contributor to **Gemini 2.5** ensuring safe and responsible multi-modal understanding and generation.
- Publishing novel research for improving safety and fairness in text-to-image models.

Apple — *Cupertino, CA*

Applied Research Engineer — *September 2020 – February 2022*

- Developing and training new models for determining the ranking of search results on the App Store.

Amazon — *Bangalore, India*

Software Development Engineer II — *October 2017 – April 2018*

- Designed and developed scalable and easy to use algorithms and platforms for solving machine learning related problems.
- Instructor for Amazon's internal machine learning bootcamps.

Software Development Engineer — *July 2015 – October 2017*

- Lead developer for EntityPredictionService (EPS), a tool for automatically generating end-to-end machine learning pipelines.

Education

Carnegie Mellon University, Pittsburgh — ***GPA: 3.95/4.0***

School of Computer Science

Master of Science in Language Technologies — *August 2018 - August 2020*

- Relevant Courses: Deep Learning, Intro to Machine Learning, Neural Networks for NLP, Intro to Optimization.
- Graduate Research Assistant

Birla Institute of Technology and Science (BITS) Pilani, India — ***GPA: 9.70/10.0***

Bachelor of Engineering (Hons) in Computer Science — *August 2011 - July 2015*

- Relevant Courses: Linear Algebra, Data Structures and Algorithms, Parallel Computing, Compiler Construction.

Publications and Pre-Prints

Going PLACES: Participatory Localized Red Teaming for Text-to-Image Safety in the Global South

Charvi Rastogi, **Mukul Bhutani**, .., Lora Aroyo

Under Review, **ACM FAccT 2026**

- Developed the PLACES dataset of 26K+ text-to-image model failures via community-centered red teaming in the Global South, uncovering novel socio-cultural harms to address contextual gaps in Western-centric AI safety frameworks.

OsCA: Benchmarking Unintended NSFW Image Generation

Susan Hao, Moonkyung Ryu, Paul Vicol, **Mukul Bhutani**, Renee Shelby, Yuqing Du, Deepak Ramachandran

Under Review, **CVPR 2026**

- Developed the OsCA benchmark of 56K benign prompts to evaluate unintentional oversexualization and intersectional biases in image generation models, assessing six mitigation strategies to improve AI safety and fairness.

Gemini 2.5

Gemini 2.5 team at Google, **Mukul Bhutani**

Technical report, 2025

- Contributor to Google's state-of-the-art large language model.
- Ensuring safe and responsible multi-modal understanding and generation.

Veo 3

Veo 3 team at Google, **Mukul Bhutani**

- Core contributor to Google's state-of-the-art video generation model.
- This is the video generation model with implicit audio generation.

Imagen 3

Imagen 3 team Google, **Mukul Bhutani**

Technical report, 2024

- Core contributor to Google's state of the art text-to-image model.
- *Blogpost link*

SeeGULL Multilingual: a Dataset of Geo-Culturally Situated Stereotypes

Mukul Bhutani, Kevin Robinson, Vinodkumar Prabhakaran, Shachi Dave, Sunipa Dev

Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics, **ACL 2024**

- Built SeeGULL Multilingual, a global-scale multilingual dataset of social stereotypes, containing over 25K stereotypes, spanning 20 languages, with human annotations across 23 regions, and demonstrate its utility in identifying gaps in model evaluations.

Harm amplification in text-to-image models

Susan Hao, Renee Shelby, Yuchi Liu, Hansa Srinivasan, **Mukul Bhutani**, Burcu Karagol Ayan, Ryan Poplin, Shivani Poddar, Sarah Laszlo

Technical pre-print, 2024

- We present and formalize the definition of *harm amplification*, a phenomenon where text-to-image models generate harmful representations that were not explicit in the input prompt.

An Empirical study to understand the Compositional Prowess of Neural Dialog Models

Vinayshekhar Kumar, Vaibhav Kumar, **Mukul Bhutani**, Alexander Rudnicky

Proceedings of the Third Workshop on Insights from Negative Results in NLP, **ACL 2022**

- Investigated generalization, syntactic robustness, and semantic capabilities of neural dialog models.

Sinkhorn-Flow: Predicting Probability Mass Flow in Dynamical Systems Using Optimal Transport

Mukul Bhutani, Thomas Magelinski and Zico Kolter

Optimal Transport Workshop, **NeurIPS 2019**

Spotlight Talk

- Predicted how discrete distributions evolve over time using optimal transport.
- Predicted the evolution of factions in social networks.

WriterForcing: Generating more interesting story endings

Mukul Bhutani*, Prakhar Gupta*, Vinayshekhar BK*, Alan W Black

Storytelling Workshop, **ACL 2019**

- Generated diverse and interesting story endings by forcing the model to attend to the keywords present in the story.

Low-rank geometric mean metric learning

Mukul Bhutani, Pratik Jawanpuria, Hiroyuki Kasai, Bamdev Mishra

Geometry in Machine Learning (GiMLi) Workshop, **ICML 2018**

- Proposed a low-rank approach to learning a Mahalanobis metric from data.
- Jointly learned a low-dimensional subspace where the data resides and the Mahalanobis metric that appropriately fits the data.

A two-dimensional decomposition approach for matrix completion through gossip

Mukul Bhutani, Bamdev Mishra

Emergent Communication Workshop, **NeurIPS 2017, Patent Granted**

- Developed a novel two-dimensional decomposition approach for matrix completion.
- The result was a setup which is a lot more secure and potentially highly parallelizable compared to traditional approaches.

MRNet-Product2Vec: A Multi-task Recurrent Neural Network for Product Embeddings

Arijit Biswas, **Mukul Bhutani**, Subhajit Sanyal

European Conference on Machine Learning **(ECML-PKDD 2017)**

- Developed a dense and low-dimensional product embedding where a diverse set of signals related to a product were explicitly injected into its representation.
- Used multimodal auto-encoder to learn language agnostic embeddings.

Skills

Languages	Python, Scala, Java, C, C++
Frameworks	PyTorch, Jax, Apache Beam, Apache Spark, TensorFlow, MATLAB